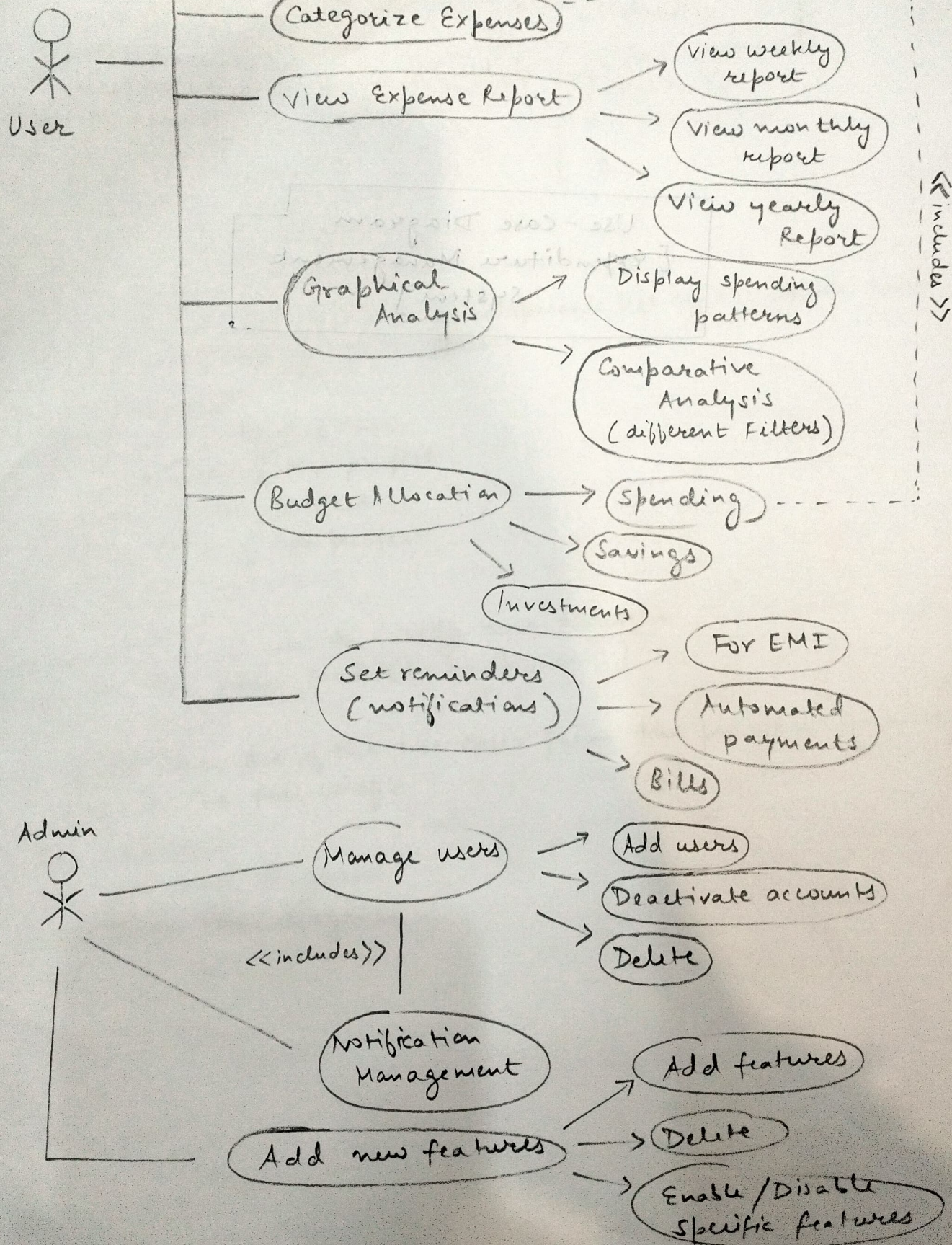
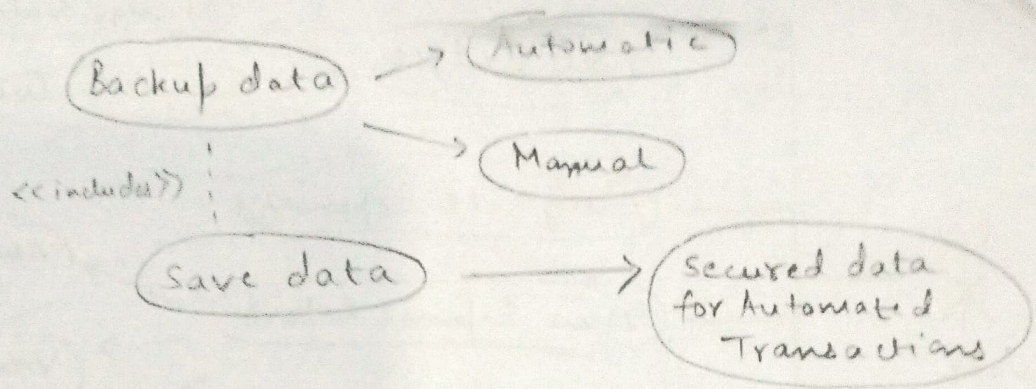


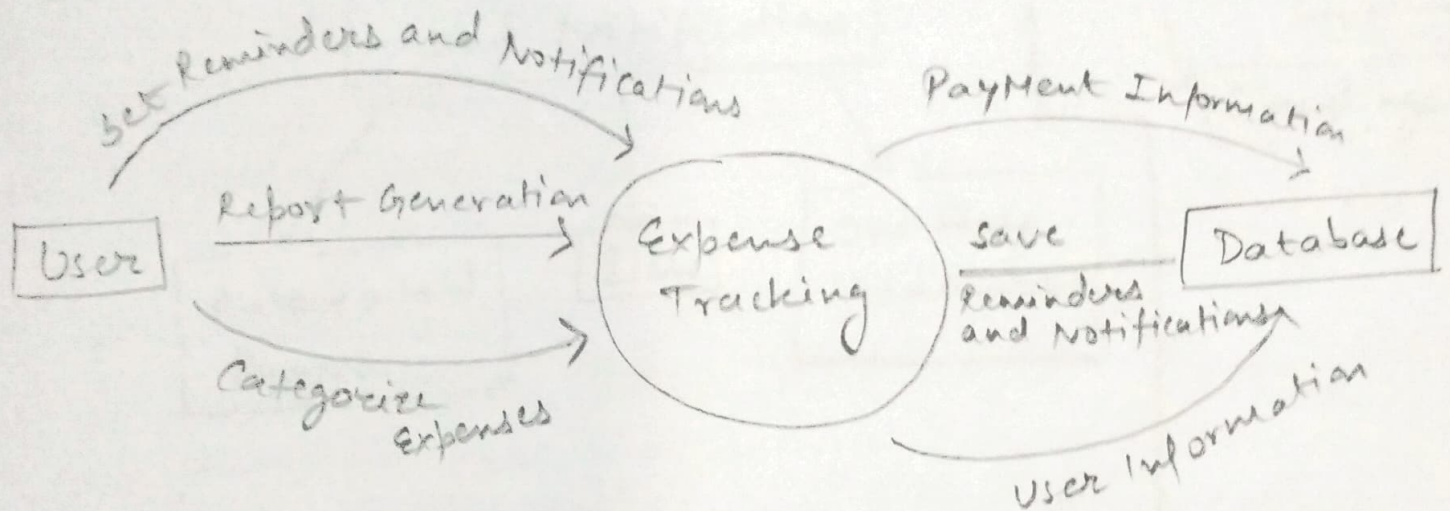


USE - CASE DIAGRAM



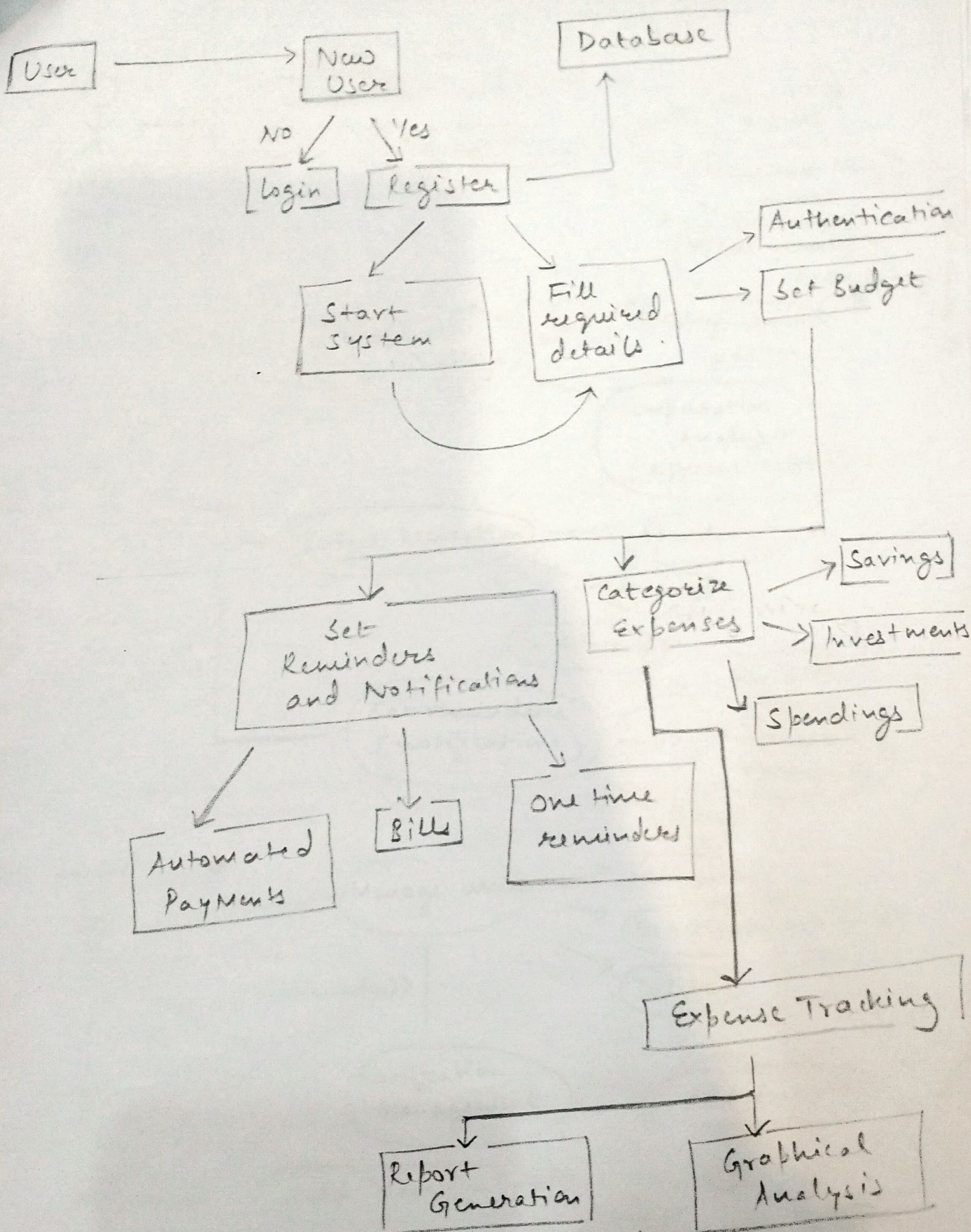
Use-Case Diagram
[Expenditure Management
System]

Data Flow Diagram [Level 0]



Data Flow Diagram [level 1]

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Question 2.

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a) SRS :- Software Requirements specification (SRS)

In Software Engineering a SRS is a Comprehensive document that outline the functional and non-functional requirements of a software system. It serves as a blueprint for developers, testers and stakeholders, ensuring a clear understanding of what the software is intended to achieve.

SOFTWARE REQUIREMENT SPECIFICATION (Expenditure Management System)

Introduction

Purpose:- The purpose of this document is to provide a Comprehensive SRS for EMS. This system is designed in a way that it aims to assist users in efficiently tracking and Managing their finances by offering functionalities such as budgeting, expense categorization, report generation and automated notifications.

Scope:- The system will provide the following functionalities:-

User Features:- Set Budget, categorize expenses, view reports (weekly, yearly, monthly), view graphical analysis and set reminders for payments.

Admin Features:- Manage users, send notifications, modify system settings, add new features and delete users.

System Features:- Automatically backup data, save records in a database and automate bill payments.

Definitions, Acronyms and Abbreviations:-

EMS :- Expenditure Management System

SRS :- Software Requirements specification

EMI :- Equated Monthly Installments

UI/UX :- User Interface / User Experience

API :- Application Programming Interface

DBMS :- Database Management System

Overall Description

Product Perspective :-

The EMS is an independent web application developed to provide users with a seamless experience in managing personal finances. It integrates with external services for functionalities like online payments and notifications, ensuring a holistic financial management solution. It also provides detailed reports to assist users in managing expenses efficiently.

Product Functions :-

1. Budgeting and Expense Management

- Set Budgets :- Users can set a budget for different expense categories such as Savings, Investments, Bills and EMI.

~~Expense Logging~~
Tracking and

- Expense logging :- Input and categorize daily expenses, attaching receipts or notes as needed.

2. Reporting and Analysis :-

- Customizable Reports :- Generate Reports based on user-defined criteria (data range, category, amount)
- Graphic Insights :- Visual tools like pie charts and bar graphs to depict spending habits.

3. Reminders and Notifications :-

- Payment Alerts :- Notifications for upcoming bills, EMIs and other financial obligations.
- Budget Alerts :- Warning when spending approaches or exceeds set budgets.

4. Administrative Controls

- User Account Management :- Admins can create, update, or delete user profiles.
- System updates :- Implements and Manage new features or system configurations
- Broadcast Notifications :- Send messages or alerts to all users regarding system updates or financial tips.

5. System operations

- Data Backup and Recovery :- Routine backup with options for data restoration.
- Automated Payments :- Schedule and process recurring payments securely.

User classes and characteristics

- End users :- Individuals seeking to monitor and manage personal finances. They require an intuitive interface with straightforward navigation.
- Administrators :- Personally responsible for overseeing system operations, user management and maintenance. They need advanced controls and access to system configurations.

Operating Environment

- Client side :- Compatible with modern web browsers (Chrome, Firefox, Safari) on desktops and mobile devices.
- Server side :- Hosted on a secure server environment, utilizing a robust DBMS (MySQL) and supporting APIs for external integrations.

Design and Implementation Constraints

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- **Security Compliance** :- Adherence to data protection regulations to ensure user information confidentiality.
- **Scalability** :- The system should accommodate a growing user base without performance degradation.
- **Third Party Integrations** :- Dependence on external APIs for functionalities like notifications and online payments, requiring consistent monitoring for changes or updates.

Assumptions and Dependencies

- Users have access to the internet and a compatible device.
- External services (eg:- payment gateways, email/SMS providers) are reliable and available.
- Users will regularly update their financial data to maintain accuracy.

Specific Requirements

Functional Requirements

1. User Module

- **Registration and Authentication**
 - Users must provide valid credentials (email, password) to create an account.
 - Implement multi factor authentication for enhanced security.
- **Budget Management**
 - Ability to set, modify or delete budgets for various categories.
 - Visual indicators to show budget utilization.

- Expense Tracking
 - log expenses with details like amount, category, date and notes.
 - upload attachments (eg - receipts) for record-keeping
- Report Generation
 - options to generate and export reports
 - Filter reports by date range, category or amount.
- Graphical Analysis
 - Display spending patterns using charts and graphs.
 - Comparative analysis b/w different time periods or categories.
- Reminders and Notifications
 - Set up one-time or recurring reminders for payments.
 - Choose notification Methods: in app alerts, email or SMS.

2. Admin Module:

- User Management
 - View a list of all users with details
 - Modify user roles or access levels.
 - Deactivate or delete user accounts as necessary.
- System Configuration
 - update system settings like default currency, time zones or language preferences
 - Manage integration with external services (eg:- payment gateways)
- Feature Management
 - Enable or disable specific features based on user feedback or requirements.
 - Deploy new functionalities with minimal disruption to users.
- Notification Management
 - Craft and send system-wide messages or alerts.

3. System Module

- Data Backup and Recovery
 - Schedule automatic backups at regular intervals
 - Provide manual backup option also
 - Ensure secure storage and easy restoration of data.
 - Automated Transactions
 - Allow users to setup recurring payments with specified frequencies.
 - Notify users before processing automated Transactions.
 - Generate system performance Analysis
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